

PRACTICAL : WEEK 7

Task - 1:

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Code:

```
#include <stdio.h>

#include <stdlib.h> int
global_var = 100; int
global_uninitialized;
void code_function()
{
    printf("Code address      : %p\n", (void *)code_function);
}
int main()
{
    static int static_var = 200;    int
stack_var = 300;    int *heap_var =
malloc(sizeof(int));

    *heap_var = 400;    printf("Linux Process
Memory Addresses\n");

    printf("-----\n");    code_function();
printf("Global variable address : %p\n", (void *)&global_var);
printf("Global uninitialized   : %p\n", (void *)&global_uninitialized);
printf("Static variable address  : %p\n", (void *)&static_var);
printf("Heap variable address    : %p\n", (void *)heap_var);
```

```
    printf("Stack variable address   : %p\n", (void *)&stack_var);  
    printf("\nProcess ID: %d\n", getpid());    free(heap_var);  
    return 0;  
}
```

Output:

```
ivikram@DESKTOP: ~/memory_lab × + ∨
ivikram@DESKTOP:~/memory_lab# nano memory_segments.c
ivikram@DESKTOP:~/memory_lab# gcc memory_segments.c -o memory_segments
ivikram@DESKTOP:~/memory_lab# ./memory_segments
Linux Process Memory Addresses
-----
Code address           : 0x5848afbfd1e9
Global variable address : 0x5848afc00010
Global uninitialized    : 0x5848afc0001c
Static variable address : 0x5848afc00014
Heap variable address    : 0x5848cf4772a0
Stack variable address   : 0x7ffd1bb0017c

Process ID: 444
ivikram@DESKTOP:~/memory_lab# |
```

Task - 2:

Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Code:

```
printf("Open another terminal and run:\n");
printf("cat /proc/%d/maps\n", getpid());
printf("\nPress Enter to terminate...\n");  getchar();
return 0;
```

```
#include <stdio.h>
#include <unistd.h> int
global_var = 100; int
main()
{
    int stack_var = 200;  printf("Process ID: %d\n",
getpid());  printf("Global variable: %p\n", (void
*)&global_var);  printf("Stack variable: %p\n", (void
*)&stack_var);  printf("\nProcess is running...\n");
```

```
}
```

Output:

```
ivikram@DESKTOP: ~  
ivikram@DESKTOP:~$ cat /proc/469/maps  
6509d79c5000-6509d79c6000 r--p 00000000 08:30 2538  
/root/memory_lab/memory_maps  
6509d79c6000-6509d79c7000 r-xp 00001000 08:30 2538  
/root/memory_lab/memory_maps  
6509d79c7000-6509d79c8000 r--p 00002000 08:30 2538  
/root/memory_lab/memory_maps  
6509d79c8000-6509d79c9000 r--p 00002000 08:30 2538  
/root/memory_lab/memory_maps  
6509d79c9000-6509d79ca000 r--p 00003000 08:30 2538  
/root/memory_lab/memory_maps  
6509eb16e000-6509eb18f000 rw-p 00000000 00:00 0  
[heap]  
750b5fa00000-750b5fa28000 r--p 00000000 08:30 42254  
/usr/lib/x86_64-linux-gnu/libc.so.6  
750b5fa28000-750b5fbb1000 r-xp 00028000 08:30 42254  
/usr/lib/x86_64-linux-gnu/libc.so.6  
750b5fbb1000-750b5fc00000 r--p 001b1000 08:30 42254  
/usr/lib/x86_64-linux-gnu/libc.so.6  
750b5fc00000-750b5fc0400000 r--p 001ff000 08:30 42254  
/usr/lib/x86_64-linux-gnu/libc.so.6  
750b5fc04000-750b5fc0600000 rw-p 00203000 08:30 42254  
/usr/lib/x86_64-linux-gnu/libc.so.6  
750b5fc06000-750b5fc13000 rw-p 00000000 00:00 0  
750b5fc38000-750b5fc3b000 rw-p 00000000 00:00 0  
750b5fc40000-750b5fc42000 rw-p 00000000 00:00 0  
750b5fc42000-750b5fc46000 r--p 00000000 00:00 0  
[vvar]  
750b5fc46000-750b5fc48000 r--p 00000000 00:00 0  
[vdso]  
750b5fc48000-750b5fc4b000 r--p 00000000 08:30 42248  
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2  
750b5fc4b000-750b5fc76000 r-xp 00001000 08:30 42248  
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2  
750b5fc76000-750b5fc80000 r--p 0002c000 08:30 42248  
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2  
750b5fc82000-750b5fc84000 rw-p 00038000 08:30 42248  
/usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2  
7fff97dc3000-7fff97dc3000 rw-p 00000000 00:00 0  
  
ivikram@DESKTOP:~$ nano memory_maps.c  
ivikram@DESKTOP:~$ gcc memory_maps.c -o memory_maps  
ivikram@DESKTOP:~$ ./memory_maps  
Process ID: 469  
Global variable: 0x6509d79c9010  
Stack variable: 0x7fff97dc10c4  
  
Process is running...  
Open another terminal and run:  
cat /proc/469/maps  
  
Press Enter to terminate...
```